Ch. 11 — P04 (K) Ranking and Selection Implementation

Problem Bank | Chapter 11 | Type: K | Difficulty:

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You are selecting the best staffing level (number of servers) for a two-stage tandem manufacturing line. The alternatives are $k = 4$ configurations:

- Config 1: (Stage 1 servers $=$ 1, Stage 2 servers $=$ 2)
- Config 2: (Stage 1 servers $=$ 2, Stage 2 servers $=$ 1)
- Config 3: (Stage 1 servers $=$ 2, Stage 2 servers $=$ 2)
- Config 4: (Stage 1 servers $=$ 1, Stage 2 servers $=$ 3)

Response: mean job sojourn time W (minutes). Parameters: arrival rate $\lambda =$ 3/min, Stage 1 mean service time 0.5 min, Stage 2 mean 0.4 min.

Tasks

1. **Pilot study:** Run $n_0 = 15$ replications (2,000 jobs each, warmup 200) for each configuration. Report the sample mean $\bar{W}_i$ and sample variance S_i^2.
2. **Naive selection:** Choose the configuration with the lowest $\bar{W}_i$. Compute a 95% CI for that configuration only. Does the CI overlap with the second-best?
3. **Rinott procedure:** Set $P^* = 0.90$ and indifference zone $\delta^* = 0.05$ minutes.
   (a) Compute the required second-stage sample size N_i for each configuration.
   (b) Run the second stage.
   (c) Report the selected best configuration and its final $\bar{W}_i$.
4. **Comparison:** Does the Rinott selection agree with the naive selection? If they differ, which do you trust more and why?
5. **Budget sensitivity:** Fill in the table below (keeping $n_0 = 15$ fixed):

δ^* (min)	P^*	Total N_i needed	Selected config
0.10	0.90		
0.05	0.90		
0.05	0.95		
0.02	0.95		

At which (δ^*, P^*) setting does the required budget first exceed 500 total replications?